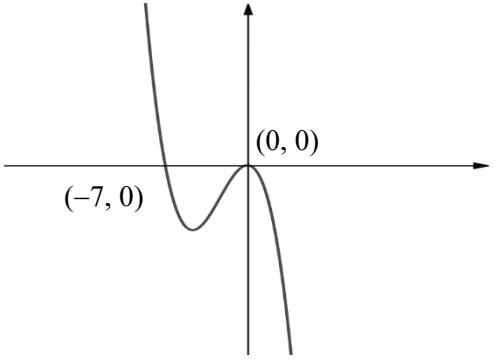
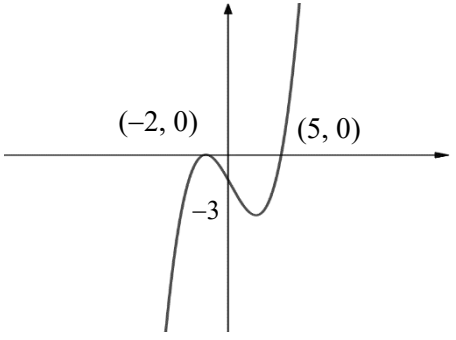
1	$a = 162$ $b = 5$ $c = 12$	B1 B1 B1
		(3 marks)

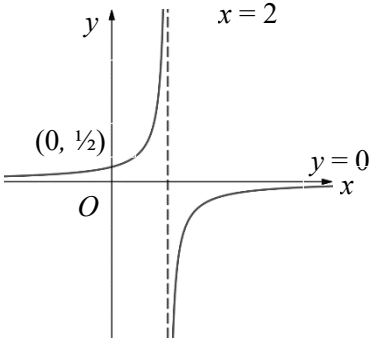
2(a)	$b = 2$	B1
	$...\pm...(x\pm 3)^2$	M1
	$\left(\mathrm{f}(x) =\right)\ 21-2(x-3)^2$	A1
		(3)
(b)	R is $(0,-4)$ or "h" $= 4$	B1
	$\mathrm{f}(x)-7=14-2(x-3)^2 \Rightarrow x=...$ or $\mathrm{f}(x)-7=-4+12x-2x^2 \Rightarrow x=...$ $\left(\text{NB } x=3\pm\sqrt{7}\right)$	M1
	$\text{Area}=\frac{1}{2}\times\left("3+\sqrt{7}"-\left("3-\sqrt{7}"\right)\right)\times"4"$	dM1
	$=4\sqrt{7}$	A1
		(4)
		(7 marks)

3(a)	$\frac{1}{2}\times 3^2\times\alpha = 7.2 \Rightarrow \alpha = ...$ or $\frac{1}{2}\times 3^2\times 1.6 = 7.2 \Rightarrow \alpha = 1.6$ $\alpha = 1.6$*	M1 A1*
		(2)
(b)(i)	Angle $COA = \frac{1}{2}(2\pi - 1.6)(= 2.34...)$ $(\approx 134°)$	M1
	Area $COA = \frac{1}{2}\times 5\times 3\sin("2.34")$ $(= 5.38...)$	M1
	Total Area $= 2\times\frac{1}{2}\times 5\times 3\sin("2.34") + 7.2$	dM1
	= 18 (cm^2) Awrt 18 (cm^2) (Ans = 17.96)	A1
(ii)	Arc $AB = 3\times 1.6(= 4.8)$	B1
	$(AC^2 =)\ 5^2 + 3^2 - 2\times 5\times 3\cos("2.34")$	M1
	Total perimeter $= 2\times\sqrt{5^2 + 3^2 - 2\times 5\times 3\cos("2.34")} + 3\times 1.6$	dM1
	= Awrt 19.6 (cm)	A1
		(8)
Alt (b)(i)	$AB = 2\times 3\sin 0.8$	M1
	$ON = 3\cos 0.8$	M1
	Total Area $= \frac{1}{2}(5 + ON)\times AB + 7.2 - \frac{1}{2}\times 3\cos 0.8\times 2\times 3\sin 0.8$	dM1
	= 18 (cm^2) Awrt 18 (cm^2) (Ans = 17.96)	A1
		(10 marks)

4	$y = 3x + 4 \Rightarrow x^2 + (3x+4)^2 + 6x - 4(3x+4) = 4$ **or** $x = \frac{y-4}{3} \Rightarrow \left(\frac{y-4}{3}\right)^2 + y^2 + 6\left(\frac{y-4}{3}\right) - 4y = 4$	M1
	$5x^2 + 9x - 2\ (=0)$ or $5y^2 - 13y - 46\ (=0)$	M1A1
	$(5x-1)(x+2) = 0 \Rightarrow x = \ldots$ or $(5y-23)(y+2) = 0 \Rightarrow y = \ldots$	dM1
	$x = 0.2,\ x = -2$ or $y = 4.6,\ y = -2$	B1
	Substitutes their x into their $y = 3x + 4$ / Substitutes their y into their $x = \frac{y-4}{3}$	M1
	$x = 0.2\left(\text{or}\frac{1}{5}\right),\ y = 4.6\left(\text{or}\,4\frac{3}{5}\,\text{or}\frac{23}{5}\right)$ **and** $x = -2,\ y = -2$	A1
		(7 marks)

5(i)(a)		B1 Horizontal translation ← B1 Maximum at origin B1 $(-7,0)$	B1B1B1
			(3)
(b)		B1 Reflection in y-axis B1 Touches at $(-2,0)$ and passes through $(5,0)$ B1 Passes through $(0,-3)$	B1B1B1
			(3)
(ii)a	$x=0 \Rightarrow y=k\cos\left(\frac{\pi}{6}\right)=\sqrt{3}$ $k\frac{\sqrt{3}}{2}=\sqrt{3} \Rightarrow k=2$		B1
(b)	$\left(p=\right)\frac{\pi}{3}$ **or** $\left(q=\right)\frac{4\pi}{3}$ $\left(p=\right)\frac{\pi}{3}$ **and** $\left(q=\right)\frac{4\pi}{3}$		B1 B1
			(3)
			(9 marks)

6(a)	E.g. $m = \dfrac{2-11}{8+4}$ or $m = \dfrac{11-2}{-4-8}$	M1
	$m = -\dfrac{3}{4}$	A1
		(2)
(b)	M is $\left(2,\ \dfrac{13}{2}\right)$	B1
	$m_N = -1 \div "-\dfrac{3}{4}"$	M1
	$y - "\dfrac{13}{2}" = "\dfrac{4}{3}"(x - "2")$	M1
	$8x - 6y + 23 = 0$	A1
		(4)
(c)	$AB = \sqrt{(-4-8)^2 + (11-2)^2}\ (=15)$ or $AB^2 = (-4-8)^2 + (11-2)^2\ (=225)$	M1
	$\dfrac{1}{2} \times MC \times AB = 37.5 \Rightarrow MC = \dfrac{75}{15}\ (=5)$ or $MC^2 = 25$	M1
	$m_N = \dfrac{4}{3}, MC = 5 \Rightarrow\ C$ is $\left("2"-3,\ "\dfrac{13}{2}"-4\right)$ or $\left("2"+3,\ "\dfrac{13}{2}"+4\right)$	dM1
	$(-1,\ 2.5)$ **or** $(5,\ 10.5)$ **or** $x = -1,\ x = 5$ **or** $y = 2.5,\ y = 10.5$	A1
	$(-1,\ 2.5)$ **and** $(5,\ 10.5)$	A1
		(5)
		(11 marks)

7(a)		B1 Negative reciprocal shape B1 Intercept at $\left(0, \frac{1}{2}\right)$ B1 $x = 2,\ y = 0$	B1B1B1
			(3)
(b)	$4x + k = \dfrac{1}{2-x} \Rightarrow (4x+k)(2-x) = 1 \Rightarrow 8x + 2k - 4x^2 - kx - 1 (= 0)$ oe		M1
	$4x^2 + (k-8)x + 1 - 2k = 0$ $a = 4,\ b = k-8,\ c = 1-2k$ or $a = -4,\ b = 8-k,\ c = 2k-1$		A1
	$(k-8)^2 - 4\times 4(1-2k) (> 0)$ oe		M1
	$k^2 - 16k + 64 - 16 + 32k > 0 \Rightarrow k^2 + 16k + 48 > 0$*		A1*
			(4)
(c)	$k^2 + 16k + 48 = 0 \Rightarrow (k+12)(k+4) = 0 \Rightarrow k = \ldots$		M1
	$k = -12,\ -4$		A1
	$k < -12$ or $k > -4$		M1A1
			(4)
			(11 marks)

8(a)	$y=(x-2)(x^2-8x+16) \Rightarrow y=x^3-8x^2+16x-2x^2+16x-32 \Rightarrow$ $y=x^3 \pm ...x^2 \pm ...x \pm 32$	M1
	$=x^3-10x^2+32x-32$	A1
	$\frac{dy}{dx}=3x^2-20x+32$ *	M1A1*
		(4)
(b)	$x=6 \Rightarrow y=(6-2)(6-4)^2=16$	B1
	$\frac{dy}{dx}=3(6)^2-20(6)+32=20$	B1
	$y-"16"="20"(x-6)$	M1
	$y=20x-104$	A1
		(4)
(c)	$3x^2-20x+32="20" \Rightarrow 3x^2-20x+12=0$	M1
	$3x^2-20x+12=0 \Rightarrow (3x-2)(x-6)=0 \Rightarrow x=...$	dM1
	$\alpha=\frac{2}{3}$	A1
		(3)
		(11 marks)

9	$\dfrac{21x^3-5x}{2\sqrt{x}}=\alpha x^{\frac{5}{2}}+\ldots$ or $\dfrac{21x^3-5x}{2\sqrt{x}}=\ldots+\beta x^{\frac{1}{2}}$	M1
	$\mathrm{f}(x)=\dfrac{27}{3}x^3-\dfrac{21}{2}\times\dfrac{2}{7}x^{\frac{7}{2}}+\dfrac{5}{2}\times\dfrac{2}{3}x^{\frac{3}{2}}(+c)\quad\left(=9x^3-3x^{\frac{7}{2}}+\dfrac{5}{3}x^{\frac{3}{2}}(+c)\right)$	M1A1A1
	$\mathrm{f}(9)=10\Rightarrow 9(9)^3-3(9)^{\frac{7}{2}}+\dfrac{5}{3}(9)^{\frac{3}{2}}+c=10\Rightarrow c=\ldots$	dM1
	$\left(\mathrm{f}(x)=\right)\ 9x^3-3x^{\frac{7}{2}}+\dfrac{5}{3}x^{\frac{3}{2}}-35$	A1
		(6 marks)